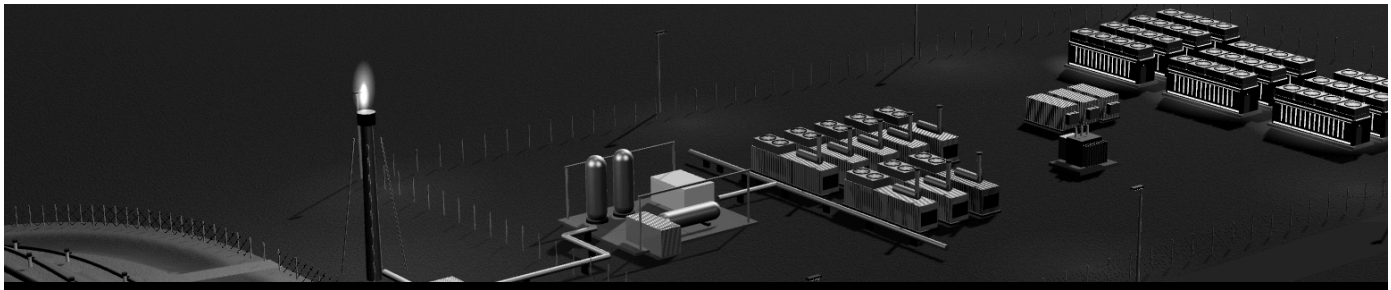
# King George Landfill, King George County, Virginia: a first sizing

What the gas the site now flares could power, how much land it would take, and what it could earn each year.

| 10.6 MW | 10 | 18 ac | 966k |
|---|---|---|---|
| POWER FROM THE GAS | COMPUTE MODULES | LAND NEEDED | MMBTU BOUGHT / YR |

| GAS SALE, PER YEAR | LAND LEASE, PER YEAR | TOTAL TO THE OWNER, PER YEAR |
|---|---|---|
| up to $2.90M | $18k to $35k | $18k to $2.93M |



YOUR SITE IN OUR LIVE 3D MODEL — ACCRETIONCOMPUTE.COM/SITE-MODEL?SRC=LFG&FLOW=3750&CH4=50.5

## THE MATH

| | |
|---|---|
| Landfill gas flared today (2023 collection less the power plant's use) | 3,750 scfm |
| Methane content (2023) | 50.5% |
| Energy in the gas | 116.1 MMBtu per hour |
| Power at a heat rate of 11,000 Btu/kWh | about 10.6 MW |
| Compute (power less cooling and losses, /1.2) | about 8.8 MW |
| Compute modules (about 1 MW each) | 10 |
| Gas engines (running plus one spare) | 8 at 1.6 MW |
| Land: 5 acres plus 1.2 acres per MW | about 18 acres |
| Gas bought per year (95% run time) | 966k MMBtu |
| Gas payment, up to $3.00/MMBtu | up to $2.90M |
| Land lease at $1,000 to $2,000 per acre per year | $18k to $35k |

Landfill gas: about 1,012 Btu per cubic foot of methane. Prices in US dollars; the price band is the one shown on our Site model page.

## WHAT IT WOULD LOOK LIKE

- Gas engines on a small parcel near the gas, a battery to smooth the load, and factory-built compute modules leased to an AI company.
- No grid connection and no export: all of the power is used on site.
- We build and run the site. The owner puts in no capital.
- Multi-year gas and land agreements, matched to the tenant's lease, at a fixed price.

### WORTH CHECKING

- Who holds the gas rights under the county host agreement
- Whether any flared gas is set aside for another project
- Gas quality (siloxanes, H2S) for engines
- Space near the landfill on the county parcel

**Source:** EPA Greenhouse Gas Reporting Program (GHGRP), facility 1007674: about 8,954 scfm collected in 2023 at 50.5% methane. The on-site power plant burned about 5,200 scfm (EIA Form 923, 2025); EPA LMOP lists 3,799 scfm flared in 2022. Sized to the flared share only. This is a first read from public data, not an offer: no cost and no obligation. The real price and terms are set in the gas and land agreements after a site review.